710001 POWER UNIT - in panel B2 (factory wired and tested; earthed aluminium enclosure)

INLET (outside B2)
powerCON TRUE1 TOP
UL 498, 20 A, locking
v1 position, X 120 Z 313

L N PE

W01
3 x 14
AWG

BREAKER 1P 15 A
UL 489, C curve;
protects the 14 AWG
mains wiring

GFCI RECEPTACLE
TR duplex NEMA 5-15R,
UL 943 Class A; unit
face, Z 1050 (D-70)

W02 L N

PE TERMINAL
inlet PE, receptacle
ground, PS1 FG, earth
stud, B2 frame bond W03

FG

W03 14 AWG green to B2's frame
(only B2 carries a mains part)

PS1 96 W CLG-100-24
kit feed, 24 V 4 A, Class 2
input 90-295 V, PFC
FG to the earth stud

F1 5 A
+24 V
0 V PS1

W02 L N

LAN
etherCON NE8FDP in B2,
W04 Cat6 to keystone 1,
keystone 2: W31 to the kit

PS2 60 W LPV-60-24
24 V 2.5 A, Class 2; light
controller, DMX512 in
Class II (no PE)

PS2 out
F2 5 A
+24 V, UP- DOWN- CALL-

RS-485

W02 L N

PS3 60 W LPV-60-12
12 V 5 A, Class 2
input 100-240 V
Class II (no PE)

+12 V, 0 V PS3

port 2

JOINTS AND HARNESS SEGMENTS (plug pair at every joint; cables clip to the cassettes only, D-26)

X1 unit / B2-CASS
X2 corner, in R1-CASS

W30b 4x16 AWG, 2 per pole B2-CASS
W30c R1-CASS

X7 VOICE KIT DROP (R1, behind the screen)
one 24 V Class 2 feed, 5 A fuse: Mini-Fit Jr 4-way, capped
Cat6 keystone, capped; draw wire W32
reached through felt plug FA-KIT

W10b B2-CASS
X4
W10c B4-CASS
X5
W11 in the shelf (6 x 18 AWG feed)

LIGHT SHELF (back and side wall tops, D-70)
uplight 4.40 m and down-light 2 x 0.95 m (24 V strips),
call drop at Y -209.5 (X6) down R2-CASS to the bar;
hub keeps up + down + call at about 22 W

W20b air feed 4x18 AWG (2 per pole)
W20c R1-CASS
W20d

W20 and W24 share one 10-way plug at X1, X2 and X3

W24b hub link 6x24 AWG: RS-485, light current, rails
W24c
W24d

W31b Cat6 (RJ45 couplers)
W31c to the kit drop X7

X3 into the air column

AIR COLUMN (front-right inside corner; hub zone 180 x 180 at Z 1200, D-61, D-70)

FUSE BLOCK
behind the hub zone cover
F3a 2 A supply fans
F3b 2 A exhaust fans
F3c 1 A hub

DESK PRO AI SENSOR HUB
in-house, cited by name; CO2, presence
out: PWM 1, 2; RS-485 to the lights
in: tach 1, 2; inlet sensor; buttons;
light current; PS1-PS3 rails (H-ELE-05)

SUPPLY FANS S1 S2
high in DR; W40 4 x 20 AWG

EXHAUST FANS E1 E2
soffit right end; W41

INLET TEMP SENSOR
supply fan box inlet; W42

CONTROLS (hub zone)
BOOST, TEST, CALL buttons
LIGHT knob: all hub inputs
(W44); TEST = self-test D-87

CALL LIGHT (R2, above the screen)
900 mm bar, 2.7 W, on light controller CH3;
fed down from the shelf (X6, W12)
see the light shelf box

SAFETY AND RESERVES
SELF-TEST (D-87): TEST checks the 3 rails, runs both fan pairs (tach), steps each light channel (current), reads the sensors and shows PASS or the failing item. PWM fail-safe: an open PWM line runs the fans at full speed. Draw wires: W32 kit drop -> column -> soffit (mic, status light); W33 back wall top -> ceiling cassette (D-41, and the sprinkler or alarm wiring point, ISS-052). Nothing is cut on site later (REQ-VCK-03).

NOTES
UNITS: mm. GENERAL TOLERANCE: harness lengths +20/-0 mm per segment; route positions +/-5 mm.
1. Mains (red) exists only inside B2 and the power unit (D-10, IF-12). Nothing mains is connected on site.
2. Every 24 V and 12 V circuit (blue) is Class 2: each supply is a Class 2 power unit on its own sheet (UL 1310), and no two meet (D-168).
3. Plug pairs: supply side receptacle (female), load side plug (male); Molex Mini-Fit Jr, keyed by way count.
4. Every cable through sheet steel runs in a grommet; cables clip to the cassette tray only (D-26).
5. Wire sizes and lengths: harness list (sheet 2 and the _SPEC.csv). Voltage drop under 3 %: calc-results.md.
6. Bought parts are named from their datasheets in the parts list 710000 (some still HOLD, each with its reason); no order, no supplier contact. The in-house hub is cited by name only.
7. Mains: US 120 V 60 Hz, 15 A, UL-listed parts (D-133); the prototype is tested in Vietnam through a 220-to-110 V step-down transformer.
8. Joints: X1 unit/B2-CASS X2 corner (R1-CASS) X3 column entry X4 B2/B4 X5 shelf X6 call X7 kit.

PROJECT	FocusPod v2 - Spaceship
TITLE	**ELECTRICAL - WIRING DIAGRAM**
DWG / PART NO.	710000 ELECTRICAL
REV / STATUS	P01 / DRAFT - not for production
UNITS / SCALE	mm / not to scale (schematic)
SHEET	1 of 3 (A1)
DATE / BY	2026-11-01 / Autonomous
SOURCE	electrical_model.py, electrical_parts.py

HARNESS LIST (every field, with conductors, routes, bushings and voltage drop: 710000_P01_ELECTRICAL-HARNESS-LIST_SPEC.csv)

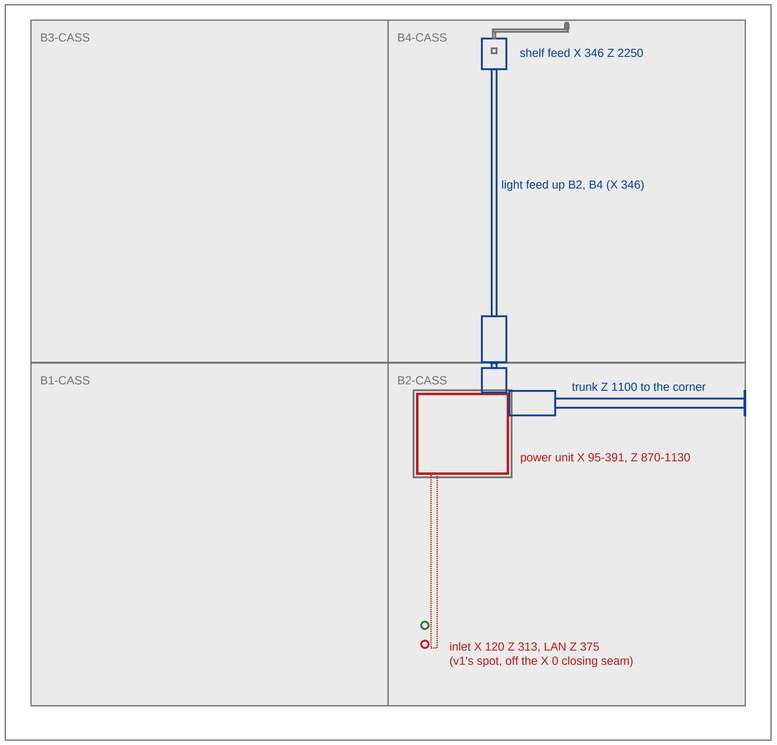
ID	SEGMENT	FUNCTION	CABLE	L mm	END A	END B
W01	PU	mains feed	UL 1015 14 AWG single cores (black, white, green) in the sleeve	920	inlet terminals (powerCON TRUE1 TOP)	breaker, PE terminal
W02	PU	mains inside the unit	UL 1015 14 AWG (black, white; green to the receptacle)	1500	breaker	GFCI receptacle LINE terminals; supply input leads in UL 486C lever connectors
W03	PU	protective bond	UL 1015 14 AWG green	300	M6 ring lug, 16-14 AWG	M6 ring lug, 16-14 AWG
W04	PU	LAN in	Cat6 U/UTP, Belden 3612 (23 AWG solid)	870	pressed into the etherCON's IDC (24-22 AWG solid)	punched down at keystone jack 1
W10a	PU	light feed	6 x 18 AWG, Class 2 (CL2 or equivalent)	300	light controller terminals	Mini-Fit Jr 6-way receptacle
W10b	B2-CASS	light feed	6 x 18 AWG, Class 2	270	Mini-Fit Jr 6-way plug (mates W10a)	Mini-Fit Jr 6-way receptacle (tail out of the top grommet)
W10c	B4-CASS	light feed	6 x 18 AWG, Class 2	1360	Mini-Fit Jr 6-way plug (mates W10b in P4)	Mini-Fit Jr 6-way receptacle, tail into the shelf
W11a	SHELF	light shelf feed lead	6 x 18 AWG, Class 2	300	Mini-Fit Jr 6-way plug (mates W10c)	strip feeds: back uplight both ways; jumpers W11b, W11c
W11b	SHELF	jumper to the left wall uplight	2 x 18 AWG, Class 2	1610	strip feed connector	strip feed connector
W11c	SHELF	jumper to the right wall: uplight, down-light, call drop	4 x 18 AWG, Class 2	1220	strip feed connector	strip feed connectors; Mini-Fit Jr 2-way receptacle for the call drop (X6)
W20a	PU	air feed	4 x 18 AWG (2 per pole), Class 2	300	PS3 output	Mini-Fit Jr 10-way receptacle (with W24)
W20b	B2-CASS	air feed	4 x 18 AWG (2 per pole), Class 2	920	Mini-Fit Jr 10-way plug (with W24)	Mini-Fit Jr 10-way receptacle (with W24), tail through the corner slot S-CNR
W20c	R1-CASS	air feed	4 x 18 AWG (2 per pole), Class 2	1260	Mini-Fit Jr 10-way plug (with W24) (mated in P2 through the felt slot FS-CNR)	Mini-Fit Jr 10-way receptacle (with W24), tail into the column
W30a	PU	voice kit feed	4 x 16 AWG (2 per pole), Class 2	300	F1, 5 A (PS1, the kit's 24 V feed)	Mini-Fit Jr 4-way receptacle
W30b	B2-CASS	voice kit feed	4 x 16 AWG (2 per pole), Class 2	920	Mini-Fit Jr 4-way plug	Mini-Fit Jr 4-way receptacle, tail through the corner slot S-CNR
W30c	R1-CASS	voice kit feed	4 x 16 AWG (2 per pole), Class 2	1050	Mini-Fit Jr 4-way plug (mated in P2 through the felt slot FS-CNR)	Mini-Fit Jr 4-way receptacle, capped with an empty plug housing (the kit outlet: one 24 V Class 2 feed, two pins per pole, 5 A fuse, 60 W peak)
W24a	PU	hub link to the power unit	6 x 24 AWG screened, Class 2	300	light controller link, PS2 current sense, PS1 and PS2 rail taps	shares W20's 10-way receptacle
W24b	B2-CASS	hub link to the power unit	6 x 24 AWG screened, Class 2	920	shares W20's 10-way plug	shares W20's 10-way receptacle, tail through the corner slot S-CNR
W24c	R1-CASS	hub link to the power unit	6 x 24 AWG screened, Class 2	1260	shares W20's 10-way plug (mated in P2 through the felt slot FS-CNR)	shares W20's 10-way receptacle
W31a	PU	voice kit data	Cat6 U/UTP	300	punched down at keystone jack 2 (unit face)	RJ45 inline coupler
W31b	B2-CASS	voice kit data	Cat6 U/UTP	920	RJ45 inline coupler	RJ45 inline coupler, tail through the corner slot S-CNR
W31c	R1-CASS	voice kit data	Cat6 U/UTP	1050	RJ45 inline coupler (in P2)	RJ45 keystone jack, capped (the kit's data link)
W20d	COLUMN	air feed	4 x 18 AWG (2 per pole), Class 2	420	Mini-Fit Jr 10-way plug (with W24d), mates W20c behind the column	fuse block input
W24d	COLUMN	hub link to the power unit	6 x 24 AWG screened	420	shares W20d's 10-way plug	hub I/O (HOLD H-ELE-05)
W40	COLUMN	supply fan pair	4 x 20 AWG screened, plenum (CMP)	1550	fuse block and hub terminals	4-way PWM fan connector into a Y-cable at the fan pair (housing HOLD H-ELE-07)
W41	COLUMN	exhaust fan pair	4 x 20 AWG screened, plenum (CMP)	1260	fuse block and hub terminals	4-way PWM fan connector into a Y-cable (housing HOLD H-ELE-07)
W42	COLUMN	inlet temperature sensor	4 x 20 AWG screened, plenum (CMP)	1750	hub sensor connector (HOLD, ISS-060)	sensor probe (HOLD H-ELE-06)
W43	COLUMN	hub supply	2 x 18 AWG, Class 2	300	fuse block	hub power connector (HOLD, ISS-060)
W44	COLUMN	controls loom	6 x 24 AWG screened	300	button and knob tags	hub inputs (HOLD H-ELE-05)
W12	R2-CASS	call light drop	2 x 18 AWG, Class 2	1250	Mini-Fit Jr 2-way plug (mates W11c's call branch in P7)	strip feed of the call bar
W32	R1-CASS + COLUMN	draw wire: kit zone to the soffit (mic, status light)	nylon draw cord 3 mm	1870	tied at P3-KIT	tied at the soffit entry
W33	B4-CASS + CEILING	draw wire: ceiling route (D-41)	nylon draw cord 3 mm	1680	tied at P6	tied at the ceiling cassette's front hole

NOTES
UNITS: mm. GENERAL TOLERANCE: harness lengths +20/-0 mm per segment; route positions +/-5 mm.
1. Lengths: route model (calculation) + 150 mm loops and tails (assumption). The kit drop has a 500 mm service loop.
2. Plug pairs: supply side receptacle (female), load side plug (male). W20 and W24 share one 10-way at X1, X2, X3.
3. Every segment is tested in the factory before it ships (D-87): continuity, polarity, no shorts, latch pull.
4. Cables: Class 2 circuits on 300 V PVC cable (CL2 or CL3 listed for the US phase); mains only inside B2 and the unit.
5. W32 and W33 are draw cords, tied off at each hole, for the voice kit's later cables (REQ-VCK-03).

PROJECT	FocusPod v2 - Spaceship
TITLE	**ELECTRICAL - HARNESS LIST**
DWG / PART NO.	710000 ELECTRICAL
REV / STATUS	P01 / DRAFT - not for production
UNITS / SCALE	mm / table, not to scale
SHEET	2 of 3 (A1)
DATE / BY	2026-11-01 / Autonomous
SOURCE	electrical_model.py, electrical_parts.py

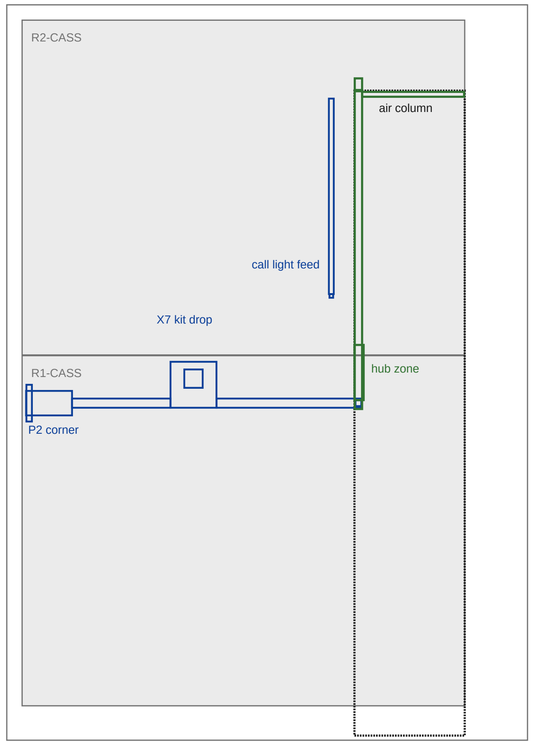
VIEW A - BACK WALL FROM INSIDE, looking toward +Y (1:10)

X view = +X, Y view = +Z; origin O on the room floor at the pod centre



VIEW B - RIGHT WALL FROM INSIDE, looking toward +X (1:10)

X view = -Y (front to the right), Y view = +Z



RESERVED SPACES IN THE CAD

Routes, pockets, bushings and slots are reserved spaces in the cassettes' 24 mm gap, in 710000_P01_ELECTRICAL_CAD, derived from skeleton P04. Pockets: P1 at the unit, P2 in R1-CASS behind the felt slot FS-CNR (hidden by B2-CASS's end), P4 above the B2/B4 joint, P5 and P6 the light feed, P3 the voice kit drop.

SITE ORDER (two people, one 5 mm key, no electrician)

1 Hang R1-CASS and R2-CASS; their tails stay out.
2 Hold B2-CASS near: plug its leads to the unit (X1) and to R1's corner tails (X2), push the pairs into P1 and P2, hang it.
3 B4-CASS: plug X4, push the pair into P4, hang it.
4 Plug the light shelf at X5 when the shelf goes up.
5 Before the air column goes in, plug X3 and X6 behind it.
6 Plug the fans at their boxes (DR inner face; soffit underside).
7 Room cord into the inlet, breaker on, press TEST: PASS (D-87).

KEEP-OUTS FOR OTHER TEAMS

WALL: B2 bay X 95-391, Z 870-1130 for the unit; sleeve X 140-160, Z 301-870; the holes and slots in B2-CASS, B4-CASS, R1-CASS, R2-CASS.
INTERIOR: ledge and screen fixings 15 mm off the trunk band Z 1085-1115.
CEILING: column chase 24 x 24 (X 1140-1164, Y -310 to -286), hub zone, soffit entry; the ceiling cassette's holes at X 583 (D-41 route).
FRONT: a 19.1 mm bushed hole in DR's back plate, X 1152, Z 2108 (G-DR).

ISO first angle

NOTES
UNITS: mm. GENERAL TOLERANCE: harness lengths +20/-0 mm per segment; route positions +/-5 mm.
1. Every coordinate in these views is computed from electrical_model.py (skeleton P04 parameters), so they follow the CAD.
2. Bushings (Heyco 2836, 19.1 hole): the unit's H01-H03, G-B2-B4 x2, G-CEIL-B/F, G-DR (DR back plate), G-SOFFIT; the slot S-CNR gets an edge strip.
3. Felt-only holes (FX-*, FS-CNR, FA-KIT) go through the felt only, clear of the 12 mm lips: no steel edge, no bushing (no hole through a lip, D-69).
4. These views are the route drawing of 710000; there is no separate Fusion drawing of the route model (PM ruling 2026-11-01).

PROJECT	FocusPod v2 - Spaceship
TITLE	**ELECTRICAL - ROUTES AND JOINTS**
DWG / PART NO.	710000 ELECTRICAL
REV / STATUS	P01 / DRAFT - not for production
UNITS / SCALE	mm / 1:10 (views)
SHEET	3 of 3 (A1)
DATE / BY	2026-11-01 / Autonomous
SOURCE	electrical_model.py, electrical_parts.py